ANADOE: Auto-Encoder-Based Network Anomaly Detection with Outlier Exposure

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***Abstract*—Anomaly detection (AD) plays a crucial role in cybersecurity by enabling the early identification of malicious activities that threaten critical systems and data. Deep Auto-Encoders (DAEs) are widely adopted for AD due to their ability to learn normal behavior in an unsupervised manner and flag deviations as anomalies. However, most existing DAE-based approaches implicitly assume that the training data are clean and benign—an assumption that rarely holds in real-world network traffic, particularly in IoT environments. When this assumption is violated, performance degrades significantly. Prior methods typically rely on hard labeling, leading to either entirely removing clusters suspected of containing anomalies or treating detected samples as final attacks. These strategies can inadvertently discard normal instances or reinforce labeling errors, especially when malicious traffic closely resembles benign behavior. To address this, we propose a DAE-based method designed to more effectively capture contamination. Our method assigns each sample a normality confidence score instead of a binary label, allowing the model to learn from contaminated data while accounting for uncertainty. Combined with outlier exposure and a weighted joint loss, this soft-labeling strategy enables the DAE to refine its decision boundary without amplifying misclassification errors. Experimental results on network traffic benchmarks, including network traffic in IoT environment datasets, demonstrate that our approach outperforms baseline methods in contaminated data settings, highlighting its effectiveness for real-world cybersecurity applications.**

***Index Terms*—Anomaly detection, IoT, Unsupervised learning, Robustness, cybersecurity, intrusion detection systems.**

I. Introduction

Anomaly detection (AD) serves as a cornerstone in contemporary cybersecurity paradigms, tasked with preemptively identifying potential security threats to safeguard critical assets and sensitive information. This pivotal function encompasses a spectrum of malicious activities, ranging from intrusion attempts and malware propagation to fraudulent behaviors [1]–[5]. In the context of cybersecurity, AD approaches are particularly relevant for intrusion detection in Internet of Things (IoT) environments [4], [5]. IoT networks, characterized by their large scale, heterogeneity, and dynamic nature, present unique challenges for intrusion detection [6]. Traditional signature-based methods are often inadequate in detecting novel or sophisticated attacks, as they rely on predefined patterns. In contrast, machine learning (ML)-based AD methods are well-suited to identifying previously unseen attacks by learning the normal behavior of IoT devices and network traffic. By leveraging ML techniques, IoT intrusion detection systems can dynamically adapt to changing patterns of normal behavior, thereby enhancing their effectiveness in identifying emerging threats. Consequently, these systems can detect various types of intrusions, including unauthorized access, data exfiltration, and denial-of-service (DoS) attacks, thus contributing significantly to the security of IoT ecosystems [6].

Among ML-based approaches, deep learning (DL)-based AD algorithms have gained significant attention due to their ability to capture complex, high-dimensional patterns within large datasets. These methods can be categorized into three main types: feature extraction, normality learning, and end-to-end anomaly score learning [7], each with distinct modeling assumptions. Feature extraction methods aim to reduce the dimensionality of the input space, employing techniques such as convolutional neural networks (CNNs) [8] to identify relevant features that make anomalies more detectable. Second, normality learning techniques, such as deep autoencoders (DAEs) [1], [9]–[11], focus on capturing the underlying regularities of normal data, under the assumption that normal instances are easier to compress or structure than anomalous ones. These models seek to reconstruct data, where high reconstruction errors indicate potential anomalies. Lastly, end-to-end anomaly score learning directly learns the distribution of anomalies and computes anomaly scores which detection decisions are based on. This approach assumes that only normal data is available during training, which is often the case in many applications where anomalies are either rare or continuously evolving, making their distribution challenging to model.

Among the array of normality learning methodologies deployed in cybersecurity [12], [13], DAEs [2]–[5] have emerged as a prevalent choice for AD within cybersecurity frameworks. These models, distinguished by their capacity to discern aberrant patterns amidst data streams in an unsupervised setting, constitute an essential component in defense mechanisms. Accordingly, our study focuses on utilizing DAEs for AD. (Other techniques will be used here for validation purposes.)

Using DAEs for AD—which aims to capture the essence of data normality or typical patterns—can significantly enhance the ability to identify deviations indicative of malicious activities. A standard assumption in this approach is that benign training data (i.e., data containing only expected, non-anomalous samples) are available for model training. However, this assumption is frequently violated in real-world scenarios [1], [14]. Training datasets are typically large and

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uncurated, and they may inadvertently contain some of the malicious instances that the model is designed to detect. This phenomenon, known as data contamination [1], [11], [15], refers to the unintentional introduction of anomalies into the training data, distinct from data poisoning—an intentional attack aimed at manipulating the model [1], [16]. For instance, data contamination may occur due to poor data quality control, equipment misconfiguration or unnoticed attack campaign. Unfortunately, contaminated training data can severely degrade the performance of DAEs [1], [15], [17] or of any unsupervised AD methods if the issue is not explicitly addressed.

In response to the challenge of contaminated training datasets, the research community has made concerted efforts to develop robust AD algorithms [1], [9], [15]. It aims to equip AD frameworks with resilience, enabling them to withstand the infiltration of malicious instances. Thus, this pursuit represents a paradigm shift toward adaptive AD methodologies capable of effectively confronting the dynamic and evolving threat landscape.

Prior methods, such as DAGMM [10] learns representations using an autoencoder trained on the normal dataset and models data likelihoods with a GMM, making it vulnerable to high anomaly contamination [9]. Other approaches attempt to separate normal and anomalous samples during training: DUAD [9] iteratively excludes suspected anomalies by hypothesizing a normal subset, while latent-label methods [15] jointly infer binary labels and model parameters, relying on outlier exposure—a training approach where a model is shown examples of out-of-distribution (unusual or irrelevant) data and is taught to respond to them with low confidence output. This helps the model better detect and handle inputs that differ from what it was trained on. However, these methods largely depend on hard assignment decisions, either permanently discarding samples or treating detected samples as definitive attacks. Such strategies are brittle when anomalies closely resemble normal behavior, as early misclassifications can be reinforced and normal data irreversibly removed. This limitation is particularly pronounced in network traffic analysis, where benign and malicious patterns are often highly overlapping, motivating the need for anomaly detection methods that can handle uncertainty without relying on irreversible labeling decisions.

Contributions. This paper introduces ANADOE, a DAE-based AD algorithm trained on contaminated datasets. Unlike hard binary labeling [15] which consists of removing potential malicious data points from the training data set, our approach addresses the issue using a soft-label approach, where each sample is assigned a confidence score of being benign, rather than relying on hard binary labels as in previous work. This soft labeling helps mitigate misclassification issues, particularly in the presence of evolving attack patterns that closely resemble benign traffic in cybersecurity and, more recently, in IoT environments. Furthermore, we leverage a Gaussian Mixture Model (GMM) to estimate the data distribution in the latent space. Also, we employ an outlier exposure mechanism, which utilizes a joint loss function to handle both benign and anomalous samples, enabling the model to learn more accurate decision boundaries even when the contamination ratio is high.

Our method demonstrates superior performance across several benchmark datasets, including CIC-IoT23, Kitsune, CIC-IDS2018R, KDDCUP, and NSL-KDD. It consistently outperforms state-of-the-art models under higher contamination levels. Ultimately, the empirical results validate the effectiveness of our approach, highlighting its practical applicability in real-world scenarios where attack samples that closely resemble benign traffic may be present in the training dataset.

II. Related Work and Preliminaries

Cybersecurity has long relied on AD to identify potential threats ranging from network intrusions to advanced persistent threats [18], [19]. In recent years, the rapid expansion of IoT networks—with their inherent scale, heterogeneity, and dynamic traffic patterns—has made traditional signature-based methods increasingly inadequate [6], [20]. IoT environments require AD systems that can adapt to evolving attack vectors while coping with diverse data sources [1], [6].

DL methods, particularly DAEs, have gained traction for modeling normal behavior in network traffic [2]–[5]. By learning compressed representations through unsupervised reconstruction tasks, these models can identify deviations indicative of potential cyber-attacks [1], [2], [17]. However, a recurring challenge in applying these methods to cybersecurity data is the assumption of clean, benign training sets. In practice, training datasets—especially those sourced from IoT devices—expected to be anomaly-free, often contain a mixture of normal and attack-related traffic due to limited data curation or covert attack strategies [1], [14], [15].

The issue of data contamination has significant consequences in cybersecurity. When malicious data inadvertently enters the training process, standard DAE approaches may learn to reconstruct anomalies effectively, thereby reducing the reconstruction error gap between normal and abnormal samples [1], [14]. In response to this challenge, recent studies have started to develop robust approaches that explicitly account for data contamination [1], [11], [15], [21], [22]. For instance, Qiu et al. [15] proposed a training strategy that infers hard binary labels for each data sample while optimizing model parameters with latent outlier exposure. This is achieved by employing a combination of two loss functions, one for normal data and another for abnormal data. Similarly, Bovenzi et al. [1] proposed enhancements to Kitnet, a two-stack ensemble of DAE architectures, to improve resilience against contamination in an IoT network environment. Meanwhile, Nkashama et al. [17] presented a latent-regulated approach based on the DAE architecture, demonstrating improvements on various traffic network datasets. Additionally, Li et al. [9] introduced a clustering-based approach that aims to remove likely anomalous data points from the training set.

In this paper, we propose a method to address the data contamination problem. Our approach is based on training a single DAE, which allows for faster inference compared to ensemble methods like that of Bovenzi et al. [1]. Our method infers soft labels rather than binary labels as in [15], [23]. Unlike [9], which removes likely anomalous data points from the training set, we reuse these identified anomalies by modifying the standard DAE loss function to incorporate an outlier exposure component. Our method demonstrates superior performance over existing approaches across several benchmark datasets with high contamination ratios, while remaining competitive in scenarios with no contamination.

A. Preliminaries

A **Deep autoencoder** is an neural network architecture that is primarily used for learning efficient data representations in an unsupervised manner. As displayed in Figure 1, it consists of an encoder network and a decoder network. The encoder network usually compresses the input data (denoted $x \in \mathbb{R}^d$) into a lower-dimensional space, also called latent representation (denoted $z \in \mathbb{R}^l$ where $l \ll d$), while the decoder network reconstructs the original input data from this compressed representation. Let's denote the encoder and decoder functions

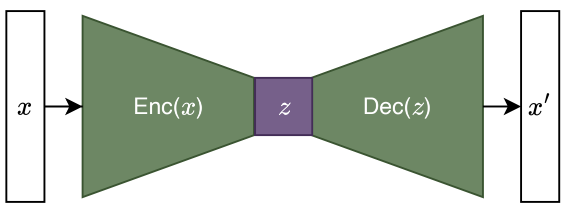


Fig. 1. DAE architecture

as enc_Θ and dec_Φ, respectively. The reconstruction of an input sample $x \in \mathbb{R}^d$ is achieved through the composition of these functions, resulting in $\hat{x} = \text{dec}_\Phi(\text{enc}_\Theta(x))$.

In the absence of any contamination in the training set, the reconstruction error, as defined in Equation (1), serves as the loss function for benign samples:

$$L_\Theta^{(n)}(x_i) = \|x_i - \hat{x}_i\|^2. \tag{1}$$

where, $\|.\|$ is the ℓ_2-norm (other loss metrics could be used).

A **Gaussian Mixture Model** is a density model [24] that estimates data distribution $p(x|\theta)$ for $x \in \mathbb{R}^d$, as a convex combination of K Gaussian distributions $\mathcal{N}(x, \mu_k, \Sigma_k)$, with $\mu_k \in \mathbb{R}^d$ the mean vector, and $\Sigma_k \in \mathbb{R}^{d \times d}$ the covariance matrix.

$$p(x|\theta) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x, \mu_k, \Sigma_k) \tag{2}$$

where $\theta := \{\pi_k, \mu_k, \Sigma_k, k = 1, \ldots, K\}$ represents model's parameters, and π_k denotes the mixture weights with $0 \leq \pi_k \leq 1$ and $\sum_{k=1}^{K} \pi_k = 1$.

GMMs serve as universal density approximators capable of representing any smooth distribution, making them uniquely suited for modeling the complex, multimodal latent spaces typically found in contaminated datasets [25]. Unlike non-parametric methods such as Kernel Density Estimation, which suffer from significant computational overhead as the number of data points and dimensions increase [26], GMMs offer a parametric efficiency that scales effectively in relatively high-dimensional latent manifolds [10].

Outlier Exposure [23] in AD consists of using an auxiliary dataset completely disjoint from the expected data in order to improve model's representation of normal data, and consequently obtain better anomaly scoring function. Let $\mathbb{P}^-$ and $\mathbb{P}^+$ denote benign (inlier) and abnormal (outlier) data distribution, respectively. The objective function with OE will then be :

$$\mathbb{E}_{x \sim \mathbb{P}^-} \left[L_\Theta(x)\right] + \lambda \mathbb{E}_{x \sim \mathbb{P}^+} \left[L_\Theta^{OE}(x)\right] \tag{3}$$

where L^{OE} denotes the loss function for outliers and $\lambda > 0$ acts as a regularization hyperparameter.

III. Method

This section outlines the problem description and our proposed method. We begin by presenting an overview of the problem, followed by a description of how our DAE-based approach leverages GMM and OE to achieve better performance.

A. Problem Description

Let us consider a dataset of network traffic samples $\mathcal{D}$, defined as

$$\mathcal{D} = \{x_i \mid i = 1, \ldots, N\},$$

where $x_i \in \mathbb{R}^d$ represents an observed network traffic sample over a specific time period. Each sample can be categorized as either benign or anomalous. Let $y_i \in \{0, 1\}$ denote the corresponding label, where $y_i = 0$ indicates benign traffic, and $y_i = 1$ indicates an anomaly. The goal is to train a model capable of accurately classifying any new, unseen sample x, regardless of its nature.

In supervised learning, both x_i and y_i are required for training. However, this approach is often infeasible in network AD due to the significant costs and effort associated with labeling data. Instead, AD techniques rely on training a model using a dataset of benign traffic,

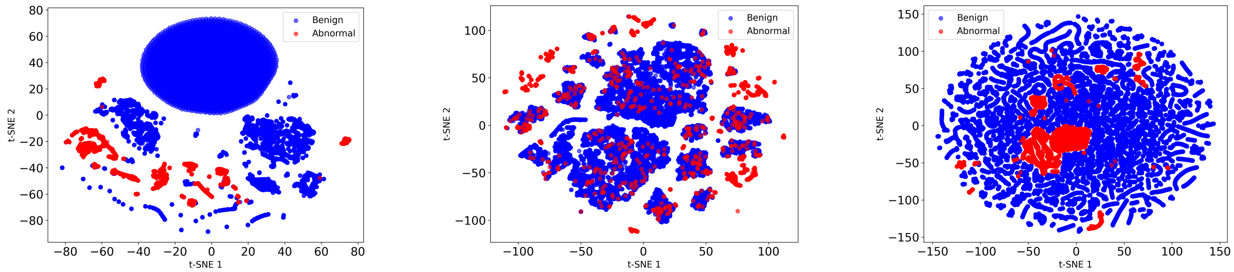
$$X^- = \{x_i \mid y_i = 0\} \subset \mathcal{D},$$

where the model learns the normal patterns of network traffic. During inference, any significant deviation from these learned patterns is flagged as an anomaly.

However, a key challenge in real-world scenarios is the assumption that the training dataset is clean—containing only benign traffic. This assumption rarely holds, as data often includes a small but non-negligible proportion of anomalies. Verification by experts to ensure clean data is both labor-intensive and error-prone, making it impractical at scale. Consequently, in real-world applications, only x is observed, while y remains largely unknown.

The presence of anomalies in the training set can significantly degrade model performance [1], [11]. For instance, a model might erroneously classify malicious traffic as benign due to its presence in the training set. This raises the critical question: *How can we train a DAE for network AD when the training dataset contains both benign and malicious traffic—a contaminated set—without access to true labels?*

To address this challenge, our objective is twofold: (1) to find means to infer y for a given x during training, and (2) to leverage detected anomaly signals from the contaminated training set to refine the anomaly scoring function. This approach aims to improve the robustness of the model and enhance its detection accuracy.



(a) KDDCUP (b) CIC-IoT23 (c) CIC-IDS2018R

Fig. 2. t-SNE visualizations of feature representations for benign (blue) and abnormal (red) samples from contaminated training datasets.

B. Proposed Approach

This section presents our proposed approach to train a robust DAE on a contaminated training dataset for detecting network anomalies. The process involves two main steps for each training iteration. Initially, we infer labels for samples without supervision signals. Subsequently, we utilize these inferred labels to train our DAE with outlier exposure, treating detected anomalies in the training set as outliers in Equation (3).

Let us denote the subset of X containing only anomalous network traffic as X^+, and its complement, $X \setminus X^+$, as X^-.

Training a DAE with outlier exposure is achieved through the loss function in Equation (4), where the objective is to minimize the reconstruction error—the loss function in Equation (1) for $x \in X^-$ while maximizing it for $x \in X^+$:

$$L_\Theta = \frac{1}{|X^-|} \sum_{x \in X^-} L_\Theta^{(n)}(x) + \frac{\lambda}{|X^+|} \sum_{x \in X^+} L_\Theta^{(a)}(x), \quad (4)$$

where

$$L_\Theta^{(a)}(x) = \frac{1}{L_\Theta^{(n)}(x) + \epsilon}, \quad \epsilon > 0. \quad (5)$$

Unlike existing approaches that infer sample labels as binary variables (0 for benign and 1 for anomalies) [15], [23], we infer labels in a continuous manner—as confidence score of a sample being benign. This choice is motivated by the evolving and sophisticated nature of attack vectors in cybersecurity, which can closely resemble benign samples [17].

In the presence of contaminated data, there is a high risk of mislabeling samples, which can significantly degrade model performance. To mitigate this, we adopt a soft-labeling strategy, where each sample is assigned a confidence score $\hat{p}_i$ of being benign and $1 - \hat{p}_i$ of being anomalous. This yields the following modified loss function:

$$L_\Theta = \frac{1}{|X|} \sum_{x_i \in X} \left(\hat{p}_i L_\Theta^{(n)}(x_i) + \lambda(1 - \hat{p}_i) L_\Theta^{(a)}(x_i) \right), \quad (6)$$

where $L_\Theta^{(n)}(x_i)$ and $L_\Theta^{(a)}(x_i)$ denote the reconstruction losses for normal and anomalous samples, respectively.

The soft-label approach offers several theoretical advantages over hard labeling in the context of contaminated datasets:

(i) ***Capturing uncertainty***. When labels are noisy or unknown, assigning binary labels ($\hat{y}_i \in \{0, 1\}$) can cause substantial errors—especially when anomalies resemble benign data (see Figure 2c). In contrast, soft labels $\hat{p}_i \in [0, 1]$ encode uncertainty directly, allowing the model to weigh both loss components adaptively. For ambiguous samples, $\hat{p}_i \approx 0.5$ balances the influence of $L_\Theta^{(n)}(x_i)$ and $L_\Theta^{(a)}(x_i)$, preventing overcommitment to incorrect assumptions.

(ii) ***Approximating expected loss***. Ideally, with known labels y_i, the loss function would be:

$$L_\Theta = \frac{1}{|X|} \sum_{x_i \in X} \left((1 - y_i) L_\Theta^{(n)}(x_i) + \lambda y_i L_\Theta^{(a)}(x_i) \right).$$

Since true labels are unavailable, we approximate the expected loss over the label distribution:

$$\mathbb{E}_x \left[P(y = 0 \mid x) L_\Theta^{(n)}(x) + \lambda P(y = 1 \mid x) L_\Theta^{(a)}(x) \right].$$

The soft-label approximates $P(y_i = 0 \mid x_i)$, aligning the training objective with the expected loss. In contrast, hard labels impose a discontinuous step-function approximation that can significantly deviate from the true expectation.

(iii) ***Mitigating misclassification impact***. Hard labels could cause the model to reinforce incorrect reconstructions—e.g., treating anomalies as benign. Soft labels, by maintaining partial weighting from both losses, reduce such risks. This flexibility enables more calibrated decision boundaries and enhances generalization, especially in scenarios with overlapping class distributions, such as modern network traffic.

Ultimately, soft labeling leads to improved learning dynamics and better performance on noisy and ambiguous data.

How is $\hat{p}$ estimated? To estimate $\hat{p}$, the input $x \in \mathbb{R}^d$ is first encoded into a latent representation $z \in \mathbb{R}^l$ using the DAE encoder, and its reconstruction $x' \in \mathbb{R}^d$ is obtained via the decoder. We then compute the reconstruction error and cosine similarity between x and x', which serve as indicators of reconstruction quality. To robustly model the data distribution, a GMM is applied. To reduce the impact of the high-dimensional input space on GMM performance, we construct a compact feature vector $v \in \mathbb{R}^{l+2}$ by concatenating the latent representation z with the reconstruction error and cosine similarity. This fused representation enhances the ability to discriminate anomalies, as shown in [9], [10]. The probability density is then estimated by:

$$\theta_{\text{GMM}} = \text{GMM}(v, K), \quad (7)$$

where K denotes the number of mixture components.

Samples located in low-density regions are more likely to be anomalous.

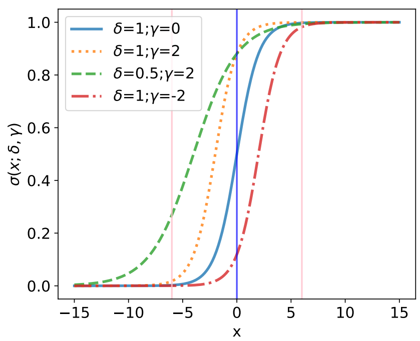


Fig. 3. Visualization of the sigmoid function in Equation (8) with different values of γ and δ.

To convert the log-likelihood from the GMM into a confidence score, we apply a sigmoid transformation:

$$\sigma_{\delta,\gamma}(x) = \frac{1}{1 + e^{-(\gamma+\delta x)}}, \tag{8}$$

where δ controls the slope and γ shifts the curve horizontally.

Since the sigmoid function saturates rapidly—flattening for large positive or small negative values—we mitigate the risk of consistently obtaining $\hat{p} \approx 0$ or $\hat{p} \approx 1$ by rescaling the log-likelihoods to lie within a target interval $[t_1, t_2]$, where the sigmoid is most sensitive. For the standard sigmoid, this interval typically spans $[-6, 6]$, as shown in Figure 3, bounded by red vertical lines. In our experiments, we set the sigmoid parameters to $\delta = 1$ and $\gamma = -2$, aiming to map the (rescaled) log-likelihoods into confidence scores in a way that respects the rarity of anomalies. Shifting the curve to the right via $\gamma = -2$ ensures that only samples with notably low likelihoods receive high anomaly scores. This design helps prevent noisy but benign samples from being misclassified, unless their scores deviate significantly. Figure 3 illustrates the effect of different parameter choices, offering guidance for future adaptations and interpretability in various contamination scenarios.

The full procedure is outlined in Algorithm 1. Figures 4a, 4b, and 4c compare the decision boundaries of DeepSVDD, a standard DAE, and our proposed method. Our approach demonstrates enhanced robustness by effectively isolating anomalies from normal data, even when trained under contamination.

GMM Estimation and Training Stability. From a theoretical perspective, the choice of GMMs in ANADOE is motivated by the interaction between representation learning and density estimation. Our DAE (Algorithm 1) is explicitly regularized to produce centered and compact latent representations, effectively constraining the latent space to a low-curvature manifold. Under such conditions, local neighborhoods of the latent space are well approximated by mixtures of Gaussian components, even when the global data distribution is complex or contaminated.

Accordingly, ANADOE performs batch-wise GMM estimation, which can be interpreted as local maximum likelihood estimation on successive samples drawn from the evolving latent distribution. While each mini-batch estimate of θ_{GMM} is biased with respect to the global density, ANADOE does not require a globally consistent density estimator. Instead, it relies on a reliable relative density ordering within each batch to identify low-density samples. This weaker requirement substantially reduces estimation variance and improves robustness in the presence of contamination.

In practice, GMM parameters are estimated independently for each mini-batch V_B (see Algorithm 1), yielding batch-wise normality estimates used for soft labeling and outlier exposure. To ensure stable estimation, we employ a sufficiently large batch size $B(2048)$, allowing each batch to capture the local structure of the latent space. Within this setting, the GMM highlights low-density regions in the batch, enabling the model to softly down-weight ambiguous or potentially contaminated samples without enforcing hard labels.

This procedure can be interpreted as a form of local outlier exposure, where robustness emerges from repeated localized density modeling rather than from fitting a single global model. By operating locally, ANADOE avoids the computational overhead and instability associated with global density estimation in high-dimensional spaces, while leveraging the parametric efficiency and analytical tractability of GMMs. The design is expected to produce stable training and greater robustness to contamination.

IV. Experiments

A. Datasets

Table I presents the statistical details of each dataset. We applied Min-Max scaling to the continuous features and one-hot encoding to the categorical features across all datasets.

TABLE I
Datasets statistics.

Dataset	# Samples	# features	Attacks ratio
Kitsune	25 758 272	115	0.2304
CIC-IoT23	4 671 520	41	0.2704
CSE-CIC-IDS2018	16 232 944	83	0.1693
KDDCUP 10	494 021	42	0.1969
NSL-KDD	148 517	42	0.4811

- **CIC-CSE-IDS2018**, created by CIC in partnership with the Communication Security Establishment, is a recent network traffic dataset that includes both normal and attack traffic on a sophisticated network setup. The dataset features various attack types, such as Brute-force, Heartbleed, Botnet, DoS, DDoS, Web-based attacks, and internal network infiltration [28].
- **Kitsune** is a cybersecurity dataset collected from an IP-based surveillance system deployed in a commercial environment. This dataset captures network traffic within an IoT setup, featuring both benign traffic and a range of cyber attacks such as OS scans, fuzzing, video injection, SSDP flooding, and Botnet malware [29].
- **CIC-IoT23** is a recently developed dataset for IoT network analysis and benchmarking. This IoT network consists of 105 devices, as described in [30]. The dataset

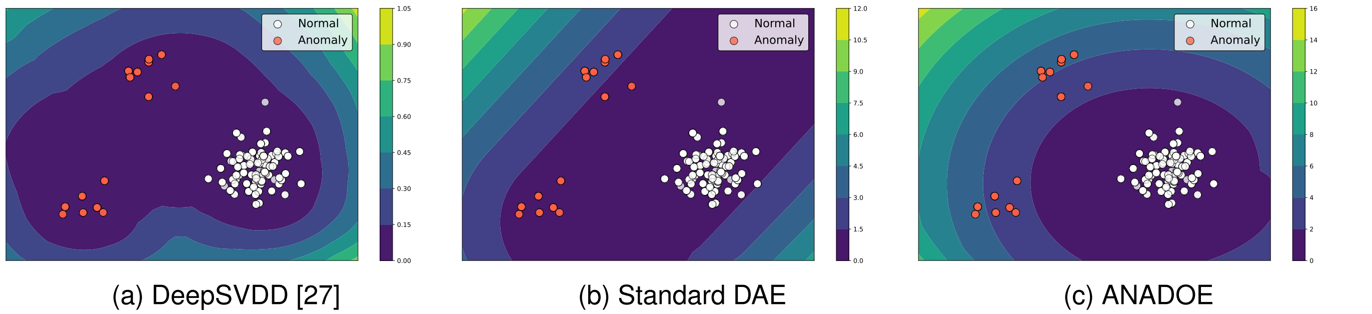


Fig. 4. The decision boundary of AD models trained on 2D synthetic and contaminated data, where white points represent benign data and red points represent malicious data.

incorporates 33 types of large-scale attacks across seven categories: DDoS (Distributed Denial of Service), DoS, Reconnaissance, Web-based, Brute Force, Spoofing, and Mirai. To better reflect real-world data distributions, we applied random downsampling on the DDoS and DoS traffic, as these attacks are generally less difficult to identify, and because their large number of instances could skew the evaluation outcomes.

- **KDDCUP**[1] is a legacy dataset for network intrusion detection, over two decades old. Despite certain limitations such as redundant samples [31], it remains widely used as a benchmark. The dataset includes both normal and attack traffic, simulating a military network with attack types like DoS, R2L, U2R, and probing. It has 41 features, of which 34 are continuous, and originally contains 80% attack data. We modified the dataset to reflect typical real-world distributions where normal data predominates, using a version that contains only 10% of the original data.
- **NSL-KDD** is a version of the KDDCUP dataset, released by the Canadian Institute of Cybersecurity (CIC). This version addresses some of the limitations of the original, such as duplicated records. Detailed information on this dataset can be found in [31].

It should be noted that we combined all attack types into a single class labeled as *"attack"* across all datasets.

B. Baselines

We implemented and investigated six anomaly detection models. Our source code can be found at https://github.com/djeffkanda/anode.

Deep Auto-Encoder. DEA is a vanilla autoencoder that maps its input data to a latent space and then attempts to reconstruct it. During training, it minimizes the reconstruction error, which is the *L2-norm* of the difference between the original input and its reconstructed counterpart.

KitNet [9]. KitNet is an ensemble method built on a series of autoencoders. It consists of k base autoencoders and a final output autoencoder. Each of the k base autoencoders maps input data to its latent space, then attempts reconstruction, producing a root mean squared error (RMSE) as the reconstruction error. These RMSEs values form a vector of dimension k, which is then processed by the output autoencoder. This final autoencoder functions as a nonlinear voting mechanism, consolidating the outputs from the ensemble.

Deep Unsupervised Anomaly Detection (DUAD) [9]. DUAD is an anomaly detection approach utilizing a DAE. It operates on the assumption that anomalous samples in the training set have a high-variance distribution. Unlike the vanilla DAE, DUAD incorporates a clustering step after a set number of iterations, isolating clusters with low variance to identify a subset of normal data. The reconstruction error determines the anomaly score.

Adversarially Learned Anomaly Detection (ALAD) [32]. ALAD builds on the Bidirectional GAN (BiGAN) architecture [33], incorporating two additional discriminators to enforce consistency between data-space and latent-space cycles. An intermediate layer's reconstruction error in one of the discriminators is used as the scoring metric.

Neural Transformation Learning for Deep Anomaly Detection Beyond Images (NTL) [15]. NTL is a self-supervised anomaly detection method that leverages contrastive learning. It combines data transformations and augmentation for tabular data using neural networks. The loss function is designed to enhance the similarity between an input and its transformations while reducing the similarity between distinct transformations of the same input. This deterministic loss function serves as the scoring mechanism. Additionally, the model incorporates a strategy for training an anomaly detector on datasets contaminated with unlabeled anomalies.

C. Evaluation Protocol

The evaluation protocol we utilize is outlined as follows (see Figure 5): initially, we divide the dataset $\mathcal{D}$ into two distinct subsets, denoted as $\mathcal{D}_1$ and $\mathcal{D}_2$. The subset $\mathcal{D}_1$ comprises a proportion r of benign samples, and $\mathcal{D}_2$ contains the remaining $(1-r)$ ratio of benign samples along with $(1-c)$ proportion of malicious samples. The remaining fraction c of malicious samples is reserved for inclusion in subset $\mathcal{C}$, which serves as a source of contamination of the training set. For a given contamination ratio α, we randomly sample $\lceil \frac{\alpha n}{(1-\alpha)} \rceil$ examples

[1] http://kdd.ics.uci.edu/databases/kddcup99/kddcup99.html

Algorithm 1 ANADOE

Require: $X = \{x_i, i = 1, \ldots, N\}$: a contaminated training set. $\mathcal{M}_\Theta = (\text{enc}_{\Theta^e}, \text{dec}_{\Theta^d})$: Deep autoencoder model. α: the learning rate. K: the number of components of the GMM. t_1, t_2: the scaling interval, the lower and upper bound, respectively. B: batch size, λ_z and λ_a: the regularization factor for the latent representation and the outlier exposure loss, respectively. T: the number of training iterations.

Initialize the model $\mathcal{M}_\Theta$ and the center $\mathbf{c}_Z$ with random weights.
for t in 1 to T **do**
  Sample $X_B = \{x_1, x_2, \ldots, x_B\}$ from X.
  Project X_B to the latent space, $Z_B = \text{enc}_{\Theta^e}(X_B)$
  $X'_B = \text{dec}_{\Theta^d}(Z_B)$, reconstruct X_B from latent representation Z_B
  Compute the losses for x_i :

$$L_i^n = \|x_i - x'_i\|^2 + \lambda_z \|z_i - \mathbf{c}_Z\|^2, \; i = 1, \ldots, B$$

$$L_i^a = \frac{1}{L_i^n + \epsilon}, \; i = 1, \ldots, B$$

$$V_B = Z_B \oplus L^n \; \oplus \text{cosim}(X_B, X'_B)$$

  Compute GMM parameters:

$$\theta_{GMM} = \text{GMM}(V_B, K)$$

  Compute samples log-likelihood for $v_i \in V_B$:

$$\ell_i = \log \sum_{k=1}^{K} \pi_k \mathcal{N}(v_i, \mu_k, \Sigma_k), i = 1, \ldots, B$$

  Compute scaled samples weight for x_i:

$$w_i = t_1 + \frac{(\ell_i - \min(\ell))(t_2 - t_1)}{\max(\ell) - \min(\ell)}, i = 1, \ldots, B$$

  Compute the final loss:

$$L = \sum_{i=1}^{N} \sigma_{\delta,\gamma}(w_i) L_i^n + \lambda_a (1 - \sigma_{\delta,\gamma}(w_i)) L_i^a$$

  Update parameters using gradient descent:
  $\Theta \leftarrow \Theta - \alpha \nabla_\Theta L$
  $\mathbf{c}_Z \leftarrow \mathbf{c}_Z - \alpha \nabla_{\mathbf{c}_Z} L$
end for

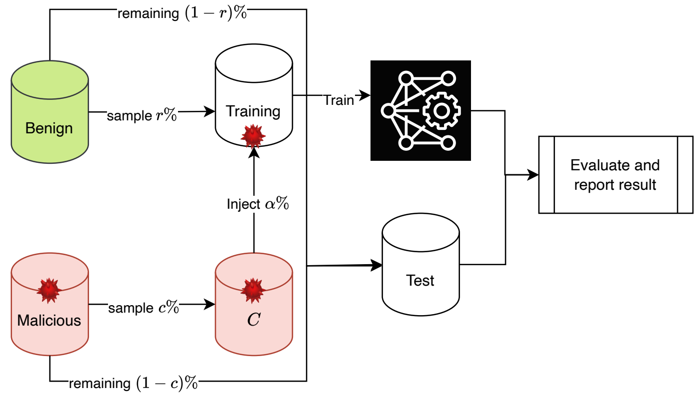


Fig. 5. Training and evaluation workflow.

from $\mathcal{C}$ to inject into $\mathcal{D}_1$, which is then utilized as the training set. Finally, we use $\mathcal{D}_2$ as the test set.

It is important to note that regardless of the contamination ratio α, the proportion of benign and malicious data in the test set are constants. This approach ensures consistent evaluation of how variations in the contamination of the training set impact model performance.

D. Experimental Setup

Training Configuration. ANADOE model was trained using the Adam optimizer with a learning rate of 10^{-3}. Training was conducted for a maximum of 200 epochs with early stopping enabled, using a patience value of 10 epochs to prevent overfitting. The GMM component was set with $K = 3$ for the KDD and NSL-KDD datasets, while $K = 7$ was used for the IoT-23, Kitsune, and CIC-IDS2018-Revised datasets.

Network Architecture. The DAE consists of an encoder-decoder structure, where d represents the number of input features in each dataset. For the KDD and NSL-KDD datasets, the encoder comprises fully connected layers of sizes (d, 180), (180, 30), (30, 20), and (20, latent_dim) with Tanh activations, except for the last layer. The decoder mirrors this structure with layers of sizes (latent_dim, 20), (20, 30), (30, 180), and (180, d), also using Tanh activations. For the IoT-23, Kitsune, and CIC-IDS2018-Revised datasets, the encoder consists of layers (d, 120), (120, 64), (64, 32), and (32, latent_dim) with ReLU activations, and the decoder symmetrically reconstructs the input using layers (latent_dim, 32), (32, 64), (64, 120), and (120, d) with ReLU activations. A simpler architecture was used for a toy dataset, where the encoder has two layers (d, 2 × d) and (2 × d, latent_dim) with ReLU activations, and the decoder symmetrically reconstructs the input. The latent dimension (latent_dim) was set to 2 for KDD and NSL-KDD datasets and 6 for the rest of the datasets.

E. Results

This section presents the performance of our network anomaly detection model on two IoT network traffic datasets—Kitsune and CIC-IoT23—as well as three additional widely used network traffic datasets: KDDCUP10, NSL-KDD,

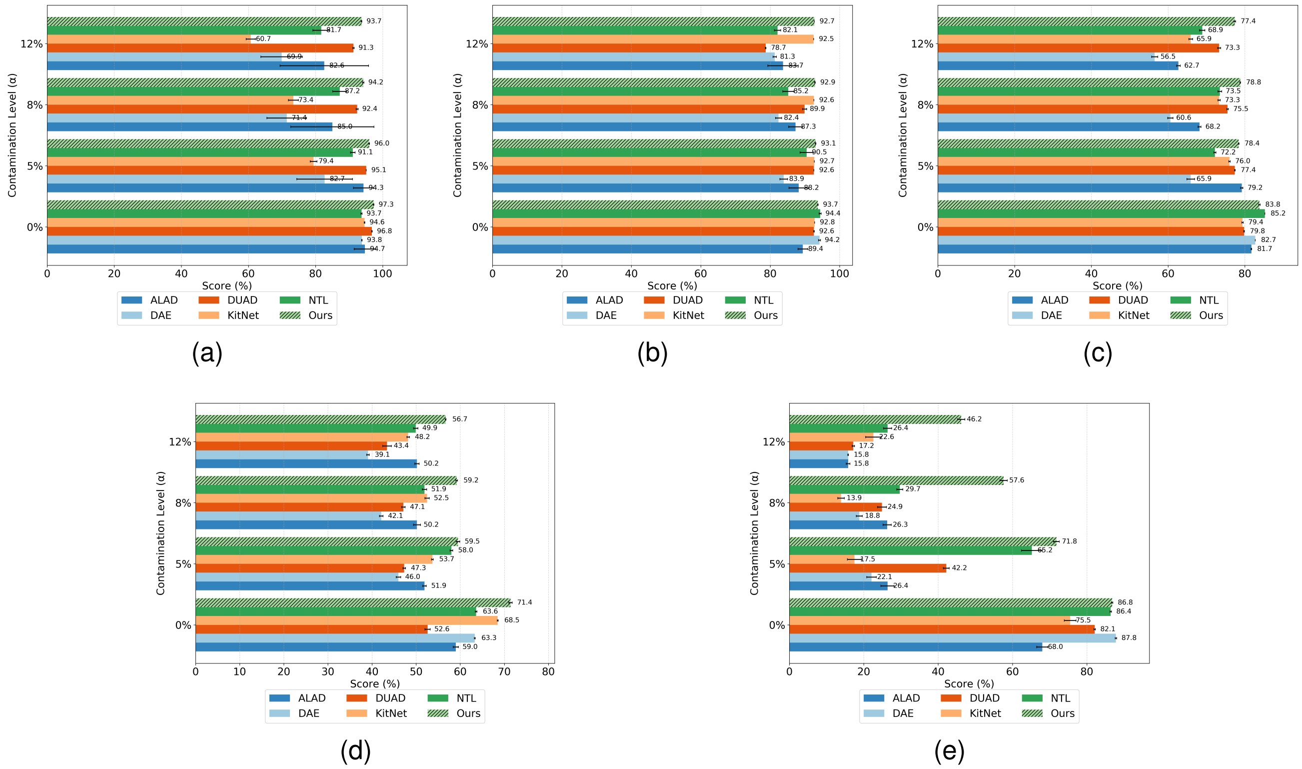


Fig. 6. F1-score–based performance comparison of models at varying contamination levels: (a) KDD dataset, (b) NSL-KDD dataset, (c) CIC-IoT23 dataset, (d) Kitsune dataset, (e) CIC-IDS2018R dataset.

TABLE II
AVERAGE PRECISION, RECALL, AND F1-SCORE (WITH STANDARD DEVIATION) OF MODELS TRAINED ON NORMAL SAMPLES MIXED WITH ATTACK SAMPLES AT A RATIO OF α.

Model	α	KDDCUP 10			NSL-KDD		
		Pr	R	F_1	Pr	R	F_1
ALAD	0%	92.5±0.9	97.2±5.5	94.7±3.2	87.1±2.5	91.9±1.1	89.4±1.4
DAE		88.8±0.4	99.4±0.4	93.8±0.1	95.3±1.7	93.2±1.2	94.2±0.3
DUAD		94.3±0.2	99.5±0.0	96.8±0.1	92.2±0.2	93.2±0.3	92.6±0.1
KitNet		89.9±0.2	99.7±0.0	94.6±0.1	91.6±0.1	93.9±0.2	92.8±0.0
NTL		88.2±0.4	99.9±0.1	93.7±0.2	95.0±0.9	93.8±0.7	**94.4±0.3**
Ours		94.9±0.2	99.9±0.2	**97.3±0.1**	92.7±0.1	94.5±0.1	93.7±0.1
ALAD	5%	90.6±4.8	98.5±1.0	94.3±3.0	88.2±2.3	88.2±4.2	88.2±2.8
DAE		79.6±7.5	86.4±10.2	82.7±8.3	82.6±2.6	85.5±4.2	83.9±1.1
DUAD		90.7±0.0	100.0±0.0	95.1±0.0	92.4±0.3	92.9±0.2	92.6±0.0
KitNet		82.8±0.5	76.7±1.3	79.4±1.0	91.9±0.1	93.7±0.2	92.7±0.0
NTL		85.1±0.5	98.0±1.4	91.1±0.7	87.0±1.8	94.3±1.9	90.5±1.9
Ours		92.4±0.2	99.9±0.1	**96.0±0.1**	89.4±0.1	97.1±0.1	**93.1±0.1**
ALAD	8%	80.5±9.6	90.6±15.8	85.0±12.4	87.3±3.9	87.5±2.2	87.3±2.0
DAE		68.5±3.8	75.1±10.2	71.4±5.9	80.2±2.4	84.8±2.8	82.4±0.9
DUAD		87.6±0.4	97.8±0.3	92.4±0.3	88.0±0.4	91.9±0.7	89.9±0.6
KitNet		75.0±1.5	71.9±1.5	73.4±1.5	91.0±0.1	94.3±0.1	92.6±0.0
NTL		81.5±2.7	94.0±2.2	87.2±2.1	82.5±1.7	88.1±2.7	85.2±1.6
Ours		89.7±0.1	99.2±0.0	**94.2±0.2**	89.5±0.1	96.5±0.1	**92.9±0.1**
ALAD	12%	77.6±9.6	89.0±17.3	82.6±13.2	80.6±4.3	87.0±4.5	83.7±4.4
DAE		69.0±7.2	71.4±8.5	69.9±6.2	79.7±1.8	83.1±1.8	81.3±0.5
DUAD		85.9±0.2	97.5±0.3	91.3±0.2	77.1±0.2	80.4±0.1	78.7±0.1
KitNet		62.0±1.3	59.5±1.6	60.7±1.4	91.5±0.1	93.6±0.1	92.5±0.0
NTL		75.4±3.2	89.4±2.8	81.7±2.5	81.2±2.7	83.2±3.0	82.1±0.9
Ours		88.3±0.2	99.9±0.1	**93.7±0.1**	89.6±0.0	95.9±0.1	**92.7±0.0**

and CIC-IDS2018R. We compare our model with several state-of-the-art approaches, including ALAD, DAE, DUAD, KitNet, and NTL. Evaluation metrics include precision (Pr), recall (R), and F1-score (F_1), with results reported alongside standard deviations to provide insights into model stability.

Performance on KDDCUP 10 and NSL-KDD.

TABLE III

AVERAGE PRECISION, RECALL, AND F1-SCORE (WITH STANDARD DEVIATION) OF MODELS TRAINED SOLELY ON NORMAL SAMPLES MIXED WITH ATTACK SAMPLES AT A RATIO OF α.

Model	α	Kitsune			CIC-IoT23			CIC-IDS2018R		
		Pr	R	F_1	Pr	R	F_1	Pr	R	F_1
ALAD	0%	63.9±0.5	54.8±0.6	59.0±0.6	87.5±0.2	76.7±0.2	81.7±0.1	63.2±1.2	73.9±1.9	68.0±1.5
DAE		63.6±0.6	63.7±0.4	63.3±0.1	89.3±0.1	77.0±0.1	82.7±0.0	86.3±0.4	89.5±0.2	**87.8±0.2**
DUAD		55.5±0.3	50.1±0.9	52.6±0.6	86.5±0.1	74.1±0.1	79.8±0.1	75.6±0.3	89.9±0.1	82.1±0.2
KitNet		72.5±0.4	65.1±0.4	68.5±0.1	81.0±0.7	78.3±0.2	79.4±0.2	72.3±1.3	79.5±1.9	75.5±1.6
NTL		69.8±0.2	58.5±0.2	63.6±0.2	90.9±0.1	80.2±0.1	**85.2±0.0**	84.5±0.5	88.6±0.2	86.4±0.2
Ours		72.2±0.7	70.7±0.5	**71.4±0.4**	90.8±0.2	77.7±0.2	83.8±0.2	80.9±0.4	93.8±0.3	86.8±0.2
ALAD	5%	52.4±0.5	52.0±0.5	51.9±0.4	84.9±0.3	74.3±0.2	**79.2±0.3**	24.1±1.6	29.2±2.0	26.4±1.8
DAE		43.5±0.4	48.9±0.6	46.0±0.5	68.3±1.0	64.1±1.1	65.9±1.0	21.9±1.2	22.5±1.4	22.1±1.3
DUAD		51.3±0.5	44.1±0.1	47.3±0.3	82.2±0.2	73.1±0.1	77.4±0.1	38.3±0.8	46.9±0.9	42.2±0.8
KitNet		58.1±0.4	50.0±0.2	53.7±0.2	77.7±0.5	74.7±0.4	76.0±0.2	17.4±1.7	18.0±2.1	17.5±1.9
NTL		60.8±0.5	55.8±0.4	58.0±0.3	69.6±0.3	75.1±0.4	72.2±0.3	62.9±2.8	67.9±2.8	65.2±2.8
Ours		55.3±0.5	64.4±0.7	**59.5±0.4**	84.2±0.4	74.4±0.2	78.4±0.1	66.0±0.8	78.7±0.9	**71.8±0.8**
ALAD	8%	53.8±1.0	47.2±0.6	50.2±0.8	66.0±0.5	70.8±0.5	68.2±0.4	24.2±1.0	28.7±1.3	26.3±1.1
DAE		39.9±0.4	44.6±0.4	42.1±0.4	63.7±0.9	57.9±0.7	60.6±0.7	19.2±0.8	18.6±0.8	18.8±0.8
DUAD		51.0±0.7	44.0±0.1	47.1±0.4	76.1±0.5	75.1±0.3	75.5±0.2	24.2±1.0	26.0±1.5	24.9±1.2
KitNet		57.0±0.7	48.8±0.4	52.5±0.5	74.2±0.3	72.8±0.6	73.3±0.3	13.9±0.8	14.1±1.1	13.9±0.9
NTL		48.7±0.5	55.7±0.5	51.9±0.5	74.8±0.5	72.7±0.7	73.5±0.5	27.7±0.7	32.3±1.0	29.7±0.8
Ours		67.1±0.4	53.0±0.3	**59.2±0.2**	84.2±0.5	74.2±0.3	**78.8±0.1**	54.7±0.7	61.6±1.4	**57.6±1.0**
ALAD	12%	50.7±0.7	50.3±0.5	50.2±0.5	61.8±0.5	63.9±0.6	62.7±0.5	14.8±0.5	17.0±0.6	15.8±0.5
DAE		37.2±0.3	41.3±0.4	39.1±0.3	59.1±0.8	54.4±0.8	56.5±0.8	15.8±0.2	15.9±0.1	15.8±0.1
DUAD		47.7±1.1	39.9±1.0	43.4±1.0	74.9±0.2	72.1±0.7	73.3±0.4	17.3±0.1	17.2±0.4	17.2±0.3
KitNet		52.2±0.3	44.7±0.3	48.2±0.3	65.7±0.3	66.5±0.9	65.9±0.5	21.9±1.9	23.5±2.4	22.6±2.1
NTL		48.1±0.7	52.1±0.4	49.9±0.5	69.9±0.8	68.6±0.9	68.9±0.7	24.4±1.0	28.8±1.2	26.4±1.1
Ours		66.4±0.6	49.4±0.2	**56.7±0.1**	80.4±0.4	74.5±0.2	**77.4±0.2**	45.4±0.9	47.9±1.3	**46.2±1.0**

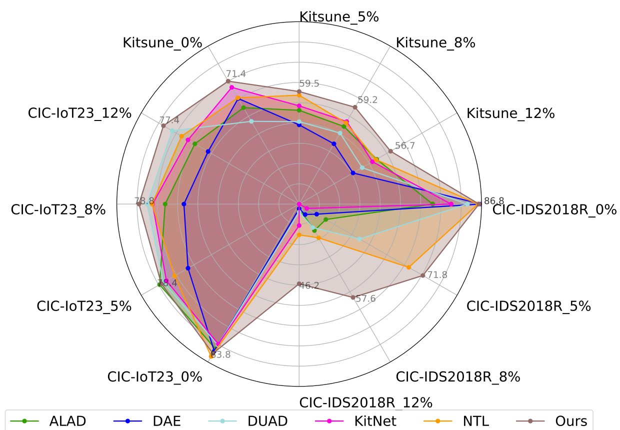


Fig. 7. Performance comparison of different models across multiple datasets (CIC-IoT23, Kitsune, and CIC-IDS2018R) using F1-scores. Each axis represents a unique dataset–contamination combination, and the overlaid curves show the performance profiles of the evaluated models.

Table II shows the performance of models trained on normal traffic data under varying contamination ratios (α) of attack samples. A summary of these results on the KDDCUP and NSL-KDD datasets, in terms of F_1-score, is displayed in Figures 6a and 6b, respectively. On both the KDDCUP and NSL-KDD datasets, our model consistently outperforms others in terms of F_1-score across all contamination levels. Specifically, at $\alpha = 0\%$, our model achieves an F_1-score of $97.3 \pm 0.1\%$ on KDDCUP and $93.7 \pm 0.1\%$ on NSL-KDD, which is comparable to top-performing models such as DUAD and NTL. Moreover, our model performs more stably than others as the contamination ratio increases.

At higher contamination levels ($\alpha = 5\%$, $\alpha = 8\%$, and $\alpha = 12\%$), the performance of our model remains stable, particularly when compared to other models, which experience significant drops in performance as attack samples increase. For instance, at $\alpha = 12\%$, our model achieves an F1-Score of $93.7 \pm 0.1\%$ on KDDCUP and $92.7 \pm 0.0\%$ on NSL-KDD. These results are significantly better than ALAD, DAE, and other baseline models, whose performance deteriorates much more rapidly as attack contamination increases.

Performance on IoT datasets: Kitsune and CIC-IoT23. Table III and Figure 7 display the performance of our model on the Kitsune and CIC-IoT23 datasets. These recent datasets exhibit more complex network traffic patterns and higher variability compared to KDDCUP and NSL-KDD. On the Kitsune dataset, our model achieves an F1-Score of $71.4 \pm 0.4\%$ at a contamination level of $\alpha = 0\%$, outperforming all other models, including KitNet, NTL, and DAE. As the contamination ratio increases, our model continues to outperform others. For example, at a contamination level of $\alpha = 12\%$, our model achieves an F1-Score of $56.7 \pm 0.1\%$, followed by ALAD, NTL, and KitNet with F1-Scores of $50.2 \pm 0.5\%$, $49.9 \pm 0.5\%$, and $48.2 \pm 0.3\%$, respectively.

On the CIC-IoT23 dataset, our model demonstrates competitive results at a contamination level of $\alpha = 0\%$, achieving an F_1-score of 83.8%, while NTL achieves 85%. As the contamination level increases, all models experience a performance drop; however, our model remains consistently less affected. For instance, at $\alpha = 8\%$ and $\alpha = 12\%$, our model outperforms others by at least 3% in F_1-score. As shown in Figure 6c, ANADOE exhibits less variation in results across runs, leading to a lower standard deviation, especially at higher contamination ratios.

Performance on CIC-IDS2018R. Figure 6e shows the models performance based on F_1-score on the CIC-IDS2018R dataset. This dataset contains network traffic data related to recent and challenging intrusion schemes as all models results testifies (details metrics with precision and recall are presented in Table III). Our model achieves a competitive result at $\alpha = 0\%$, with an F_1-score of $86.8 \pm 0.2\%$, similar to the performance of NTL, while DAE achieves $87.8 \pm 0.2\%$. However, a significant difference in performance emerges as the contamination ratio increases. For instance, at $\alpha = 5\%$, our model achieves an F_1-score of $71.8 \pm 0.8\%$, outperforming all other models, with NTL following at $65.2 \pm 2.8\%$. At a higher contamination ratio (i.e., $\alpha = 12\%$), our model achieves an F_1-score of $46.2 \pm 0.0\%$, outperforming other models by at least 20%.

Generally, all models perform poorly when tested on the contaminated version of this dataset. This led us to further investigate the reasons for these unsatisfactory results—specifically, why every model achieved an F_1-score below 50% under contamination.

As shown in Figures 2a, 2b, and 2c, the t-SNE visualizations highlight significant differences in data distribution across datasets. In Figure 2c, corresponding to the CIC-IDS2018R dataset, the red points (anomalies) are densely interspersed among the blue points (benign data), with no clear boundary between the two classes. This indicates a high degree of overlap in the feature space. In contrast, Figure 2a shows red points forming distinct clusters, clearly separated from the blue region, making anomalies easier to detect. Figure 2b presents a noisier distribution, yet still exhibits some separation between red and blue points. However, in Figure 2c, the red points are entirely surrounded by blue ones, forming a complex, entangled manifold. The t-SNE projection reveals that anomalous samples in this dataset do not form distinct low-dimensional embeddings, which increases model confusion.

In other words, the attacks resemble benign behavior so closely that the models struggle to learn effective decision boundaries in this highly non-linearly separable setting, when the training set is contaminated. This observation is supported by clustering quality metrics. The silhouette score (range: $[-1, 1]$) measures the cohesion and separation of clusters, with higher values indicating more well-defined clusters. The Calinski-Harabasz score assesses the ratio of between-cluster dispersion to within-cluster dispersion, with higher scores indicating more distinct clusters.

The CIC-IDS2018R dataset shows a silhouette score of 0.047 and a Calinski-Harabasz score of 813, indicating weak cluster separation and high intra-class overlap. In contrast, KDD10 shows a silhouette score of 0.25 and a Calinski-Harabasz score of 5896, suggesting that anomalies in that dataset are more structurally distinct and easier to learn. These metrics quantitatively support the t-SNE visual analysis and explain the poor F_1-scores in contamination settings, particularly for the CIC-IDS2018R dataset.

ANADOE vs. DAE hard-labeling. It is crucial to compare hard and soft labeling when using DAE to demonstrate the benefits of employing soft labels, as applied in our model. Figures 8a, 8b, 8c, and 8d compare the F_1-score performance of our proposed method, ANADOE, against a DAE with hard labeling (DAE_HL) across varying contamination levels. The results consistently show that ANADOE outperforms DAE_HL across all datasets and contamination settings.

The key theoretical advantage of ANADOE lies in its use of soft labels to assign a confidence score of benignness to each sample, rather than forcing binary decisions. By doing so, the model can learn more nuanced decision boundaries and avoid overfitting to incorrect assumptions. This approach mitigates the risks of mislabeling during training—especially in the presence of label noise caused by contamination. This is particularly important in modern network traffic, where the distinction between benign and malicious behavior is often subtle. As illustrated in Figure 2c, samples associated with attacks can closely resemble those associated with normal activity. In such cases, hard labeling may falsely treat contaminated samples as definitively benign, misleading the model during training. Soft labeling allows the learning algorithm to remain cautious—implicitly reflecting uncertainty—thus improving robustness. This is especially beneficial under realistic conditions with limited or noisy labels, making ANADOE more suitable for handling contaminated datasets, as confirmed by its superior empirical performance.

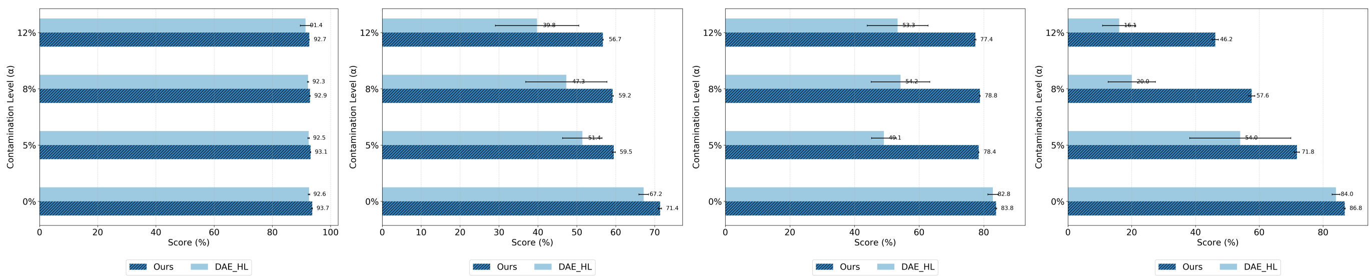
F. Ablation Study

Compared to the standard DAE, ANADOE introduces three additional components, each expected to positively influence model performance. To validate this assumption, we conduct an ablation study. Table IV analyzes the contribution of each component of the proposed ANADOE framework across three network intrusion datasets (Kitsune, CIC-IoT23, and CIC-IDS2018R) under increasing levels of training data contamination (0%, 8%, and 12%). Results are reported as mean F1-score $\pm$ standard deviation.

As discussed before, the standard DAE exhibits performance degradation as contamination increases across all datasets, confirming its strong reliance on clean training data. This effect is particularly pronounced on CIC-IDS2018R, where the F1-score collapses under moderate contamination, highlighting the limitations of reconstruction-based anomaly detection in realistic network traffic.

Incorporating probabilistic latent modeling (DAE_GMM—our first additional component) improves performance only when the level of contamination is negligible, thus it remains highly sensitive to contamination with high variance when it is run multiple times. Adding latent centering (DAE_CR_GMM—our first two additional components) yields consistent improvements over DAE_GMM, indicating that constraining the latent space better facilitates separation between benign and anomalous samples. However, performance still degrades as contamination increases.

ANADOE, which integrates latent centering, GMM and outlier exposure with a soft-labeling strategy (outlier exposure and soft labeling are our third additional component), consistently achieves the highest F1-scores across nearly all datasets



(a) NSL-KDD (b) Kitsune (c) CIC-IoT23 (d) CIC-IDS2018R

Fig. 8. Comparison of F1-score performance between the DAE hard-labeling method and ANADOE under different contamination levels.

TABLE IV
ABLATION STUDY: F1-SCORE COMPARISON (MEAN ± STD) ACROSS DATASETS AND CONTAMINATION.

Model	Kitsune			CIC-IoT23			CIC-IDS2018R		
	0%	8%	12%	0%	8%	12%	0%	8%	12%
DAE	63.3 ± 0.1	42.1 ± 0.4	39.1 ± 0.3	82.7 ± 0.0	60.6 ± 0.7	56.5 ± 0.8	**87.8** ± **1.7**	18.8 ± 7.6	15.8 ± 1.0
DAE_GMM	52.5 ± 6.1	47.1 ± 3.8	43.5 ± 10.4	79.6 ± 1.1	74.4 ± 2.3	71.9 ± 2.4	82.1 ± 1.9	24.9 ± 12.2	17.2 ± 2.5
DAE_CR_GMM	67.7 ± 0.8	50.8 ± 2.9	49.0 ± 3.9	**87.4** ± **1.2**	77.9 ± 2.8	71.4 ± 3.0	83.1 ± 4.3	44.7 ± 4.7	41.3 ± 1.1
ANADOE	**71.4** ± **0.4**	**59.2** ± **0.2**	**56.7** ± **0.1**	83.8 ± 0.2	**78.8** ± **0.1**	**77.4** ± **0.2**	86.8 ± 0.2	**57.6** ± **1.0**	**46.2** ± **1.0**

and contamination levels considered here. Notably, the performance gap widens as contamination increases, demonstrating that soft labeling enables the model to leverage contaminated samples without amplifying misclassification errors. These results validate the effectiveness of ANADOE for anomaly detection in realistic IoT traffic environments.

TABLE V
HYPERPARAMETER SENSITIVITY USING WALD TESTS (MIXED-EFFECTS MODEL). (β^* REPRESENTS THE STANDARDIZED EFFECT SIZE).

Parameter	Coef. (β^*)	Std. Err.	z-value	p-value
Intercept	0.731	0.061	11.91	$< 0.001^{***}$
Main Effects				
K	0.005	0.002	3.01	$\mathbf{0.003}^{**}$
δ	0.001	0.002	0.71	0.480
γ	0.000	0.002	0.21	0.834
Interaction Effects				
$K \times \alpha$	-0.007	0.002	-3.75	$< \mathbf{0.001}^{***}$
$\delta \times \alpha$	0.001	0.002	0.49	0.625
$\gamma \times \alpha$	-0.004	0.002	-2.32	$\mathbf{0.020}^{*}$

Significance codes: $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$

G. *Hyperparameter Sensitivity and Statistical Significance*

Recall that ANADOE relies on three main hyperparameters (K, δ, γ), so assessing its robustness to variations in these parameters is essential. To do so, we fit a mixed-effects model where the three hyperparameters and the contamination ratio (α) are treated as fixed effects, while each dataset as a random intercept (see results in Table V).

a) Stability of δ and γ: The analysis reveals that ANADOE is robust to variations of δ ($\beta^* = 0.001$, $p = 0.48$) and the parameter γ ($\beta^* \approx 0.000$, $p = 0.83$). This indicates that a light fine-tuning of these hyperparameters could lead to near-optimal results.

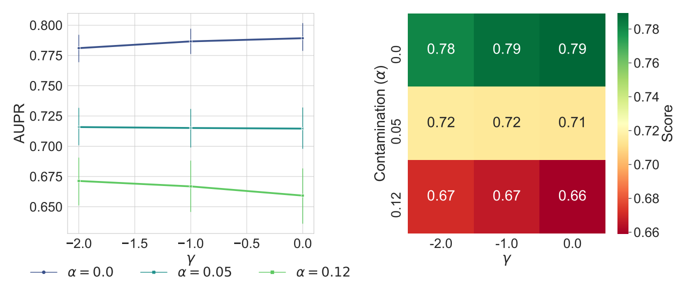


Fig. 9. Interaction between the hyperparameter γ and the contamination ratio.

Figure 9 shows interactions between γ and the contamination ratio. Under low contamination, performance is stable across γ, consistent with its negligible effect when examined alone. However, under higher contamination levels, increasing γ leads to a modest performance decrease, explaining the significant but weak $\gamma \times \alpha$ interaction observed in the mixed-effects sensitivity analysis.

b) Impact of Components Count (K): The number of components K shows a statistically significant but modest positive effect on performance ($\beta^* = 0.005$, $p < 0.01$). As illustrated in Figure 10, the effect of the number of components is modulated by contamination: increasing K is beneficial when α is small, but can be harmful at higher contamination levels, confirming the significant interaction identified by the mixed-effects model. Post-hoc Kruskal-Wallis tests confirm that this sensitivity is dataset-dependent: it is significant for the CIC-IoT23 dataset ($p_{\text{adj}} < 0.05$), suggesting that more complex data manifolds benefit from a higher K.

Overall, the sensitivity analysis demonstrates that ANADOE is robust to hyperparameter selection and exhibits controlled, interpretable behavior. Performance is primarily driven by the contamination ratio α, reflecting dataset difficulty rather than

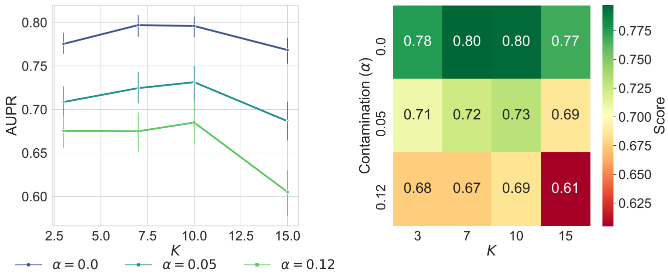


Fig. 10. Interaction between the number of components K and contamination ratio.

tuning instability. Among hyperparameters, only the number of components K shows a modest but statistically significant influence, whose effect is further modulated by contamination level. In contrast, δ and γ remain stable across datasets and contamination levels, substantially reducing the need for extensive hyperparameter tuning.

H. Discussion

Our results demonstrate the effectiveness of our network AD model across multiple datasets with varying levels of attack contamination. A key observation is that our model consistently outperforms others, achieving higher F1-Scores even at higher contamination levels ($\alpha = 8\%$, $\alpha = 12\%$) while remaining competitive when there is no contamination.

One of the main strengths of our model is its ability to identify attack-related samples in the training set as the contamination ratio increases. Unlike models such as DUAD, our approach reuses the identified outlier samples in the loss function to refine the learned decision boundary and enhance the anomaly scoring function. Experimental results show that other models experience drops in F1-Score at higher contamination levels, particularly on recent datasets, whereas our model remains less affected. This suggests that our approach is robust to data contamination. Compared to baseline models, it is well-suited for real-world applications, where attack samples may closely resemble benign traffic and go unnoticed in the training set.

Furthermore, our model's ability to handle the diverse nature of IoT traffic is particularly noteworthy. In the case of CIC-IoT23, which involves heterogeneous IoT devices with varying communication patterns, our model achieves the highest F1-Score at higher contamination levels ($\alpha = 8\%$, $\alpha = 12\%$), while remaining competitive at $\alpha = 0\%$. This demonstrates its potential for deployment in IoT networks, where traffic patterns differ significantly from traditional network traffic, and where traditional methods may struggle to generalize effectively.

Limitations. While ANADOE is designed to operate under contaminated training data, its soft-labeling mechanism is optimized for scenarios where anomalous samples are present during training. When trained on fully clean data, the estimated normality confidence scores may be less discriminative than those of methods explicitly optimized for clean settings; nevertheless, ANADOE remains competitive across the evaluated benchmarks. In addition, ANADOE relies on a GMM to characterize the latent distribution, and its performance can be sensitive to the choice of the number of mixture components K. As observed in the sensitivity analysis, certain datasets benefit from dataset-specific tuning of K. Incorporating non-parametric alternatives, such as a Dirichlet Process GMM, into the training loop represents a promising direction to eliminate this dependency and improve adaptability. Also, ANADOE assumes that the proportion of benign samples dominates the training data, which is consistent with most real-world IoT traffic scenarios. In extreme cases where anomalous traffic is equally prevalent or dominant, the model's ability to establish a reliable normal reference may degrade. Addressing such highly contaminated settings remains an open challenge.

V. Conclusion and Future Work

In this paper, we presented ANADOE, an AD approach based on DAE trained on a contaminated dataset. By leveraging a soft-label framework, this approach allows the DAE to infer, during training, confidence score that a sample is benign or anomalous. The proposed model utilizes OE, where detected anomalies are treated as outliers during training, enhancing the decision boundary and improving network AD accuracy. We also incorporated a GMM in the latent space to better capture the underlying distribution of the data and refine the confidence estimation, which is crucial for distinguishing between benign and attack traffic, especially in high-contamination scenarios.

Our empirical results across several datasets—including KDDCUP, NSL-KDD, Kitsune, and CIC-IoT23—demonstrated that our model mostly outperforms existing methods in terms of F1-Score, achieving the best performance at high contamination levels ($\alpha = 12\%$). The model maintained a consistent performance across different contamination ratios, suggesting that it is well-suited for real-world deployment in cybersecurity settings where attack traffic closely resembles benign behavior and may be found in the training set.

Our approach offers an improved solution for network AD in contaminated data environments. The combination of a soft-label approach, OE, and GMM-based density estimation provides a powerful tool for detecting anomalies in network traffic, making it a promising candidate for deployment in dynamic and adversarial environments such as IoT. Future work will focus on optimizing the model's efficiency, extending its applicability to other domains, and further enhancing the label inference process to accommodate even more complex attack scenarios. While ANADOE operates unsupervised, its flexible soft-labeling framework can integrate traditional signature-based methods for enhanced anomaly detection. For instance, known attack signatures could inform the soft-labeling process by setting dynamic thresholds for confidence scores or by providing prior knowledge to refine the estimation of $\hat{p}_i$. This hybrid approach could significantly enhance the model's ability to definitively identify faulty or compromised nodes, especially in scenarios where certain attack patterns are well-characterized. Such integration would seamlessly combine the strengths of unsupervised learning—its robustness to contaminated data and adaptability to novel attacks—with the precision of signature-based methods for known threats, thereby making ANADOE suitable for a wider range of cybersecurity applications.

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